Area, Surface Area & Volume

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Sine, cosine, and tangent ratios

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

2 [Foundational] • Target: Sine, cosine, and tangent ratios

A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$.

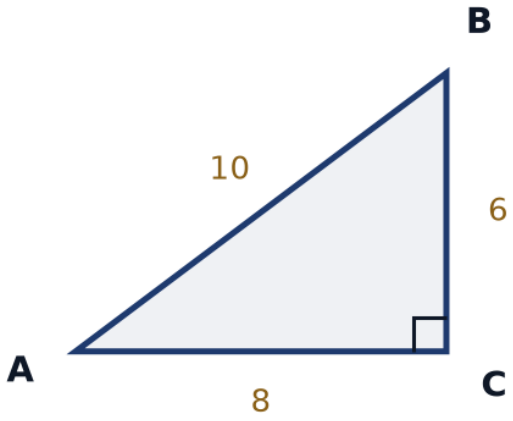
What is the arithmetic mean? Show your complete algebraic work.

3 [Foundational] • Target: Sine, cosine, and tangent ratios

For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(3)$.

4 [Foundational] • Target: Sine, cosine, and tangent ratios

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



5 [Foundational] • Target: Sine, cosine, and tangent ratios

For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(4)$.

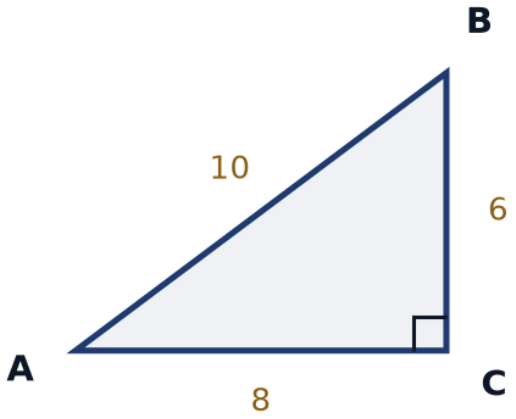
6 [Foundational] • Target: Sine, cosine, and tangent ratios

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

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7 [Foundational] • Target: Sine, cosine, and tangent ratios

For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(5)$.



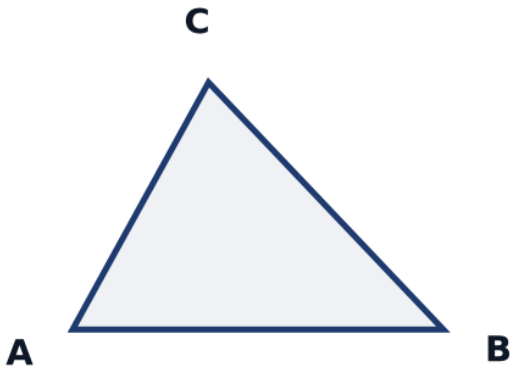
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sine, cosine, and tangent ratios

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

9 [Intermediate] • Target: Solving for sides in right triangles

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 [Intermediate] • Target: Solving for sides in right triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

11 [Intermediate] • Target: Solving for sides in right triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

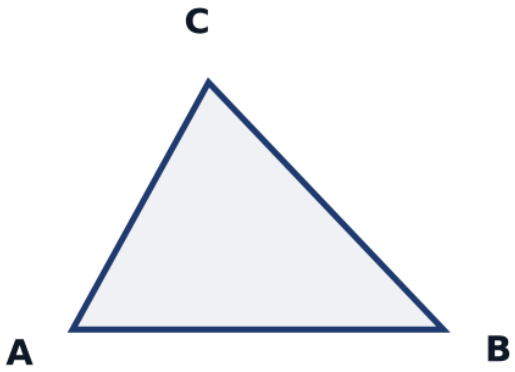
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12 [Intermediate] • Target: Solving for sides in right triangles

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

13 [Intermediate] • Target: Solving for sides in right triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.



14 [Intermediate] • Target: Solving for sides in right triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 2.01 Lines & Angles

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

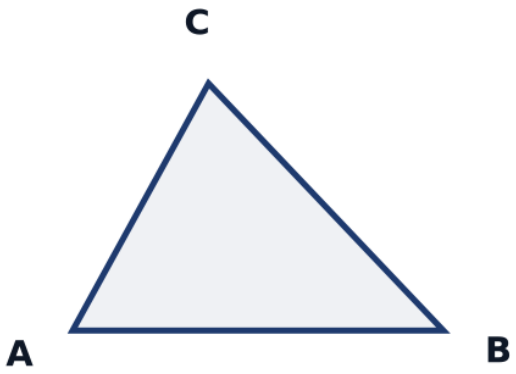
16 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

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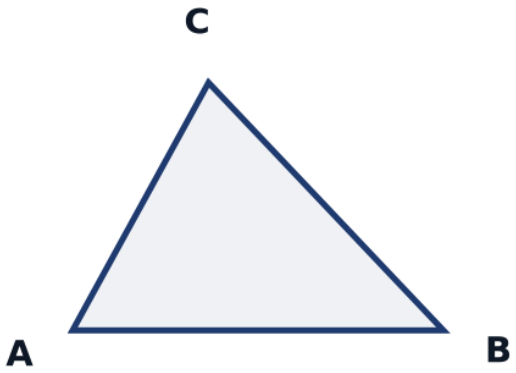
17 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



18 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



19 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A geometric figure scaled by factor $k = 5$ has an original side length of $4\ \text{cm}$. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 7.01 Circles

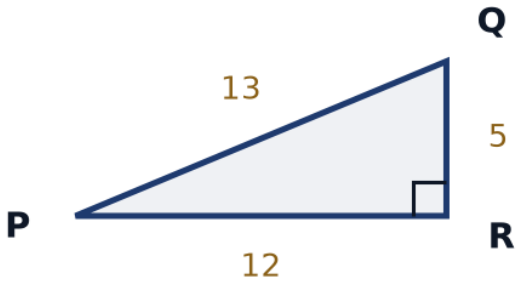
On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 88°

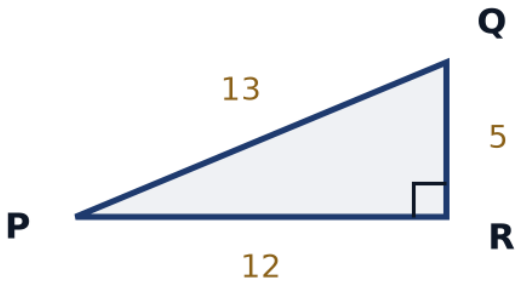
b. 98°

c. 80°

d. 133°

22 [Foundational] • Target: The Pythagorean theorem and its converse

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

b. 18 cm

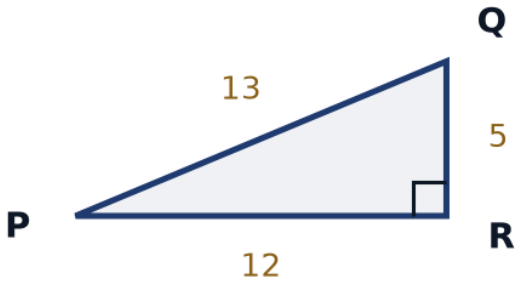
c. 24 cm

d. 54 cm

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23 [Foundational] • Target: The Pythagorean theorem and its converse

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?



a. 11

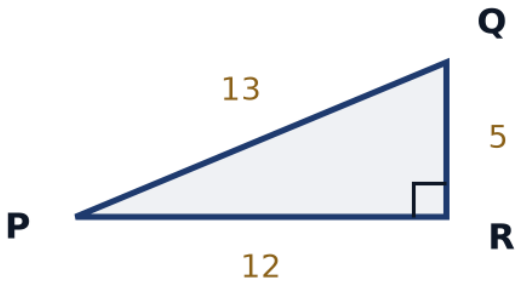
b. 17

c. 15

d. 19

24 [Foundational] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 67°

b. 77°

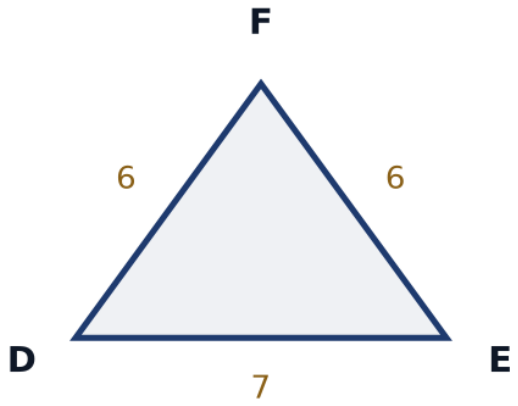
c. 59°

d. 133°

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25 [Foundational] • Target: Extension: Special right triangles

In ΔABC studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 87°

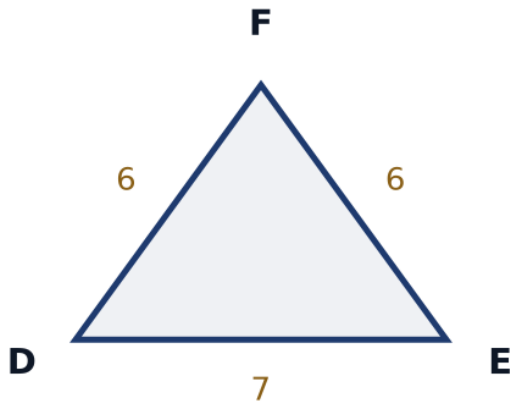
b. 97°

c. 79°

d. 132°

26 [Foundational] • Target: Extension: Special right triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?



a. 18 cm

b. 21 cm

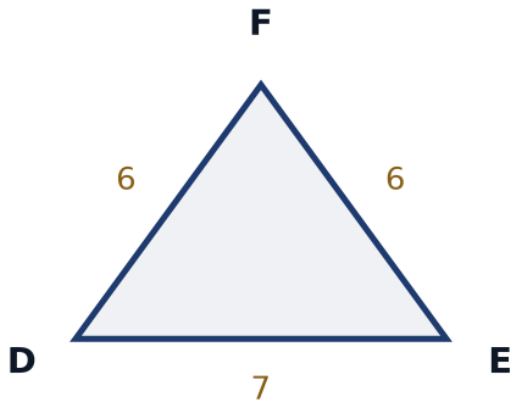
c. 28 cm

d. 63 cm

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27 [Foundational] • Target: Extension: Special right triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?



a. 14

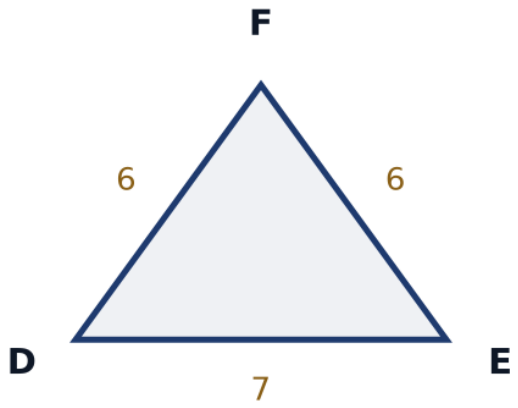
b. 21

c. 19

d. 24

28 [Foundational] • Target: Extension: Special right triangles

In ΔABC studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 66°

b. 76°

c. 58°

d. 132°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: The Pythagorean theorem and its converse

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

30 [Intermediate] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 81°

b. 91°

c. 73°

d. 133°

31 [Intermediate] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 74°

b. 84°

c. 66°

d. 133°

32 [Intermediate] • Target: The Pythagorean theorem and its converse

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

33 [Intermediate] • Target: Extension: Special right triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

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34 [Intermediate] • Target: Extension: Special right triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 9

b. 16

c. 14

d. 19

35 [Intermediate] • Target: Extension: Special right triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

36 [Intermediate] • Target: Extension: Special right triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 19

b. 26

c. 24

d. 29

37 [Intermediate] • Target: Solving for sides in right triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$?

a. 9

b. 14

c. 12

d. 15

39 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

40 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 7.01 Circles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 15

b. 20

c. 18

d. 21

41 [Challenging] • Target: Solving for angles in right triangles • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 13

c. 11

d. 15

42 [Challenging] • Target: Solving for angles in right triangles • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70°

b. 80°

c. 62°

d. 129°

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43 [Challenging] • Target: Solving for angles in right triangles • Cross-Lesson: 10.01 Coordinate Geometry

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

44 [Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 11.01 Probability

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 15

c. 13

d. 18

45 [Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 13

b. 20

c. 18

d. 23

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